STATGR 5261 STATISTICAL METHODS IN FINANCE



Portfolio Optimization For Common ETFs

Final Project Report

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1. Introduction

Portfolio optimization aims to balance return, risk, and diversification when allocating capital across assets. In this project, we compare four strategies on a common universe of 15 liquid ETFs: an equal-weight benchmark, mean-variance optimization, minimum-variance optimization, and CVaR optimization. To make the comparison more realistic, we evaluate rolling-window versions of the optimization-based strategies using only information available at each decision point. Our goal is to compare these methods in terms of return, volatility, drawdown, tail risk, and turnover, and to examine the trade-offs between simplicity, stability, and optimization.

2. Data Description

A daily multi-asset return dataset is used for portfolio optimization and out-of-sample backtesting, covering October 21, 2021, to April 17, 2026, with 1,126 observations and 17 columns, including the date, daily risk-free rate, and returns for 15 assets. The investment universe covers major asset classes such as equities, bonds, commodities, real estate, U.S. dollar exposure, and Bitcoin futures. The risk-free rate is based on FRED's DTB3 3-month Treasury bill rate, converted to a daily rate and aligned with ETF trading dates. The final dataset contains no missing values and supports excess return and risk-adjusted performance analysis. For backtesting, a 252-day rolling window is used: at each rebalancing point, portfolio inputs are estimated using only the most recent 252 trading days, and the optimized weights are applied to the next out-of-sample period, allowing the portfolio to adjust over time while avoiding look-ahead bias.

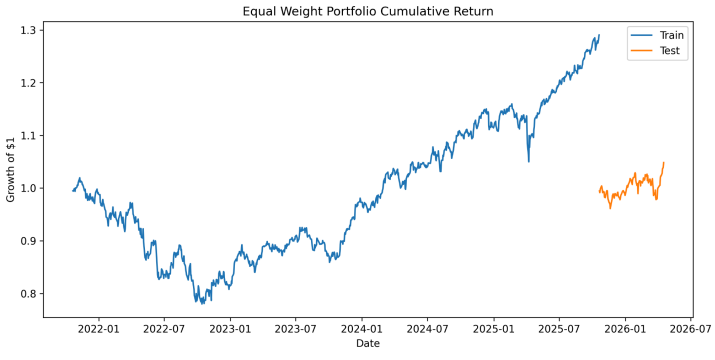
3. Methodology

3.1 Equal-Weighted Portfolio

The equal-weight portfolio is used as a simple benchmark strategy. Instead of optimizing portfolio weights, this method allocates the same weight to all 15 ETFs. Since there are 15 assets in the portfolio, each asset receives a weight of: $w_i = 1/15$. This benchmark provides a useful comparison for the more advanced optimization models discussed later in the report. Because all assets are treated equally, the strategy avoids estimation error from expected returns or covariance matrices. Using the daily ETF return data, we calculate the portfolio return as the weighted average of all asset returns:

$$R_{p,t} = \sum_{i=1}^{15} w_i R_{i,t}$$

The excess return is then obtained by subtracting the daily risk-free rate from the portfolio return: $R_{p,t}^{excess} = R_{p,t} - R_{f,t}$



To evaluate portfolio performance, several standard performance metrics are computed, including mean daily return, daily volatility, Sharpe ratio, cumulative return, and maximum drawdown. The Sharpe ratio is calculated using excess returns: Sharpe Ratio = E(Rp − rf) / σ(Rp − rf).

The cumulative return curve is also plotted to show the growth of one dollar invested in the equal-weight portfolio over time. Compared with optimized portfolios, the equal-weight strategy is simple and stable because it does not rely on parameter estimation or frequent rebalancing. However, it may not fully utilize differences in risk and expected return across assets. Therefore, this benchmark is mainly used to evaluate whether more advanced portfolio optimization methods can generate better risk-adjusted performance.

3.2 Mean-Variance Optimization

3.2.1 Static Mean-Variance Optimization

We employ the Mean-Variance Optimization (MVO) framework based on Modern Portfolio Theory. Using the 2021-2025 training set, we estimate the annualized expected return vector μ and covariance matrix $\mathbf{\Sigma}$. Our objective is to find the portfolio weights ω that maximize the exante Sharpe Ratio: $\max_w \frac{w^T \mu - r_f}{\sqrt{w^T \Sigma w}}$

This is solved using the Sequential Least Squares Programming (SLSQP) algorithm, subject to the full investment ($\sum \omega i = 1$) and long-only ($0 \leq \omega i \leq 1$) constraints. We also map the Efficient Frontier by minimizing portfolio volatility across 50 target returns.

3.2.2 Dynamic Extensions: Rolling Window and GARCH

To account for time-varying financial data, we evaluate two dynamic MVO approaches on the out-of-sample test set using a 252-day moving window with monthly rebalancing:

1. **Rolling MVO:** Continuously re-estimates μ and $\mathbf{\Sigma}$ using the trailing window to adapt to recent market regimes.
2. **GARCH Dynamic MVO:** To model volatility clustering, we apply a GARCH(1,1) process to forecast conditional variance: $\sigma^2_{i,t} = \omega_i + \alpha_i \epsilon^2_{i,t-1} + \beta_i \sigma^2_{i,t-1}$, where α_i and β_i capture short-term shocks and long-term persistence, respectively. We then construct the dynamic covariance matrix as $\mathbf{\Sigma}_t = D_t R_t D_t$, where D_t is a diagonal matrix of GARCH-forecasted volatilities ($\sigma_{i,t+1}$) and R_t is the rolling sample correlation matrix. This allows the optimizer to swiftly adapt to varying risk environments.

3.3 Minimum Variance Portfolio

We first construct a static minimum-variance portfolio on the training sample by minimizing portfolio variance under full-investment and long-only constraints. Let Σ denote the sample covariance matrix of asset returns. The portfolio weights w are obtained by solving

$$\min_w w^\top \Sigma w \text{ subject to } \sum_{i=1}^N w_i = 1, w_i \geq 0$$

for all i. This static specification provides a baseline allocation implied by the historical covariance structure. To capture changing volatility and correlation patterns across assets, we extend the static framework using a rolling-window specification. Specifically, at each rebalance date, we estimate the covariance matrix from the previous 252 trading days and form a new minimum-variance portfolio, which is then held for the next 21 trading days. Unlike the static version, the rolling strategy allows the portfolio to respond to time-varying risk structure in the data, so that the allocation can adjust as recent co-movement and asset-level volatility evolve over time.

3.4 CVaR Optimization

We first construct a static CVaR portfolio on the training sample by minimizing historical 95% CVaR under full-investment and long-only constraints. Let r_t denote the vector of daily asset returns at time t, and let w denote the portfolio weight vector. The static CVaR portfolio is obtained by solving

$$\min_w \ \mathrm{CVaR}_{0.95}\left(-r_t^T w\right) \quad \text{subject to} \quad \sum_{i=1}^N w_i = 1, \ w_i \geq 0$$

for all i. This static specification provides a baseline allocation implied by historical downside-risk minimization.

To account for time-varying tail risk, we then extend the static framework using a rolling-window specification. Specifically, we estimate the portfolio weights from the previous 252 trading days and rebalance the portfolio every 21 trading days during the test period. At each rebalance date, a new rolling-window CVaR portfolio is formed and then held until the next rebalance date. Compared with the static version, the rolling strategy allows portfolio weights to adjust dynamically as downside-risk conditions evolve over time.

4. Result Discussion

4.1 Equal Weight Portfolio

	Mean Return (daily)	Volatility (daily)	Sharpe Ratio (daily)	Cumulative Return	Max Drawdown
Train	0.00027882802066895600	0.007003645832723300	0.017990169167573600	0.2905473335493440	-0.2350044385698380
Test	0.00040394056689693100	0.006252826205486800	0.04152045557510630	0.04842613348846500	-0.049590700518171800

equal-weight performance summary table

The equal-weight portfolio achieves positive cumulative returns in both the training and test periods. During the training period, the portfolio generated a cumulative return of about 29.05%, with a Sharpe ratio of approximately 0.018. However, the maximum drawdown reaches around -23.50%, which shows that the portfolio experienced a relatively large decline during some periods.

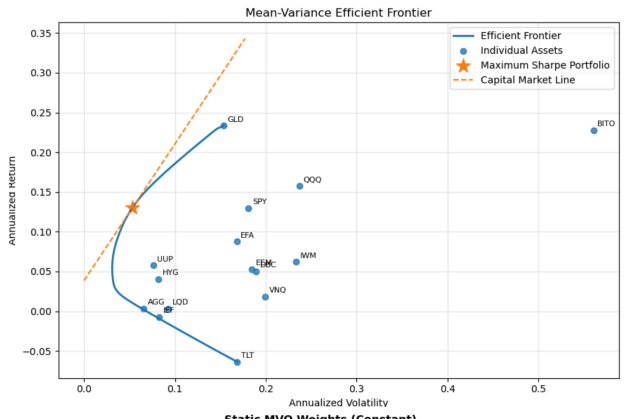
In the test period, the cumulative return is about 4.84%. Although the return is smaller than the training period, the portfolio shows lower volatility and a much smaller maximum drawdown of around -4.96%. The Sharpe ratio also increases to about 0.0415, indicating improved risk-adjusted performance out of sample. The cumulative return figure shows that the portfolio experienced fluctuations during 2022 and early 2023, followed by a gradual upward trend afterward. In the test period, the portfolio remains relatively stable and continues to generate positive returns.

Overall, the equal-weight strategy provides stable and diversified exposure across all ETFs. Even though it does not optimize weights based on expected return or covariance estimation, it still produces reasonable performance with relatively controlled risk during the test period. Therefore, the equal-weight portfolio serves as an important benchmark for comparing the performance of more advanced portfolio optimization strategies later in the report.

4.2 Mean-Variance Optimization

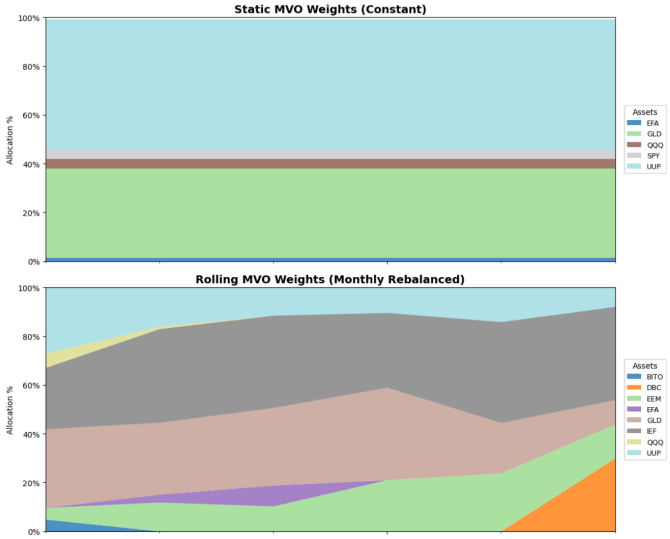
4.2.1 In-Sample Optimization and the Efficient Frontier

Using in-sample estimates, the static MVO maps the efficient frontier and identifies the Maximum Sharpe Ratio portfolio, shown by the orange star. This point represents the tangency between the efficient frontier and the Capital Market Line from the risk-free rate, illustrating the optimal historical risk-return trade-off.

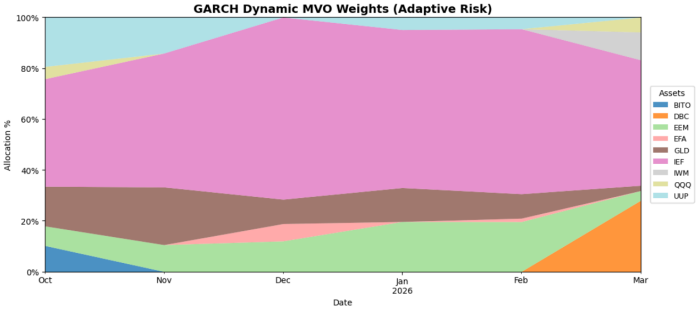


4.2.2 Portfolio Weight Dynamics and Turnover

The weight charts illustrate clear differences in allocation behavior across the three strategies. Static MVO produces fixed horizontal bands, implying no rebalancing and a concentrated "corner solution" in historical winners such as GLD and UUP. Rolling MVO updates allocations using a 252-day window, improving diversification but introducing higher turnover. GARCH Dynamic MVO is the most active strategy, shifting aggressively toward safe-haven assets such as IEF when forecasted volatility rises. To reduce excessive trading, the GARCH variance forecast is smoothed over 21 days, keeping average turnover at 0.187, only slightly above Rolling MVO at

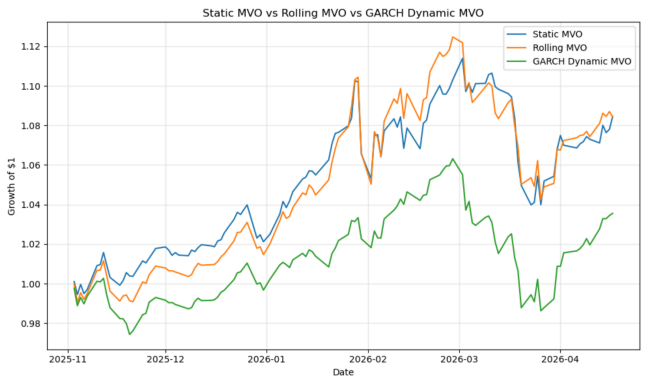


0.158. However, this risk-responsive mechanism also creates a conservative tilt that limits upside potential.



4.2.3 Out-of-Sample Evaluation: Static vs. Dynamic Approaches

Out-of-sample results show that dynamic modeling did not outperform Static MVO. Rolling MVO produced slightly lower returns, higher volatility, and a lower Sharpe ratio, suggesting that frequent parameter re-estimation introduced estimation noise. GARCH Dynamic MVO achieved the lowest volatility and drawdown, but also the lowest Sharpe ratio, as its conservative allocation reduced risk without improving return forecasts. Overall, turnover control helped limit transaction cost concerns, but the dynamic strategies failed to generate superior risk-adjusted performance during the test period.



4.3 Minimum Variance Portfolio

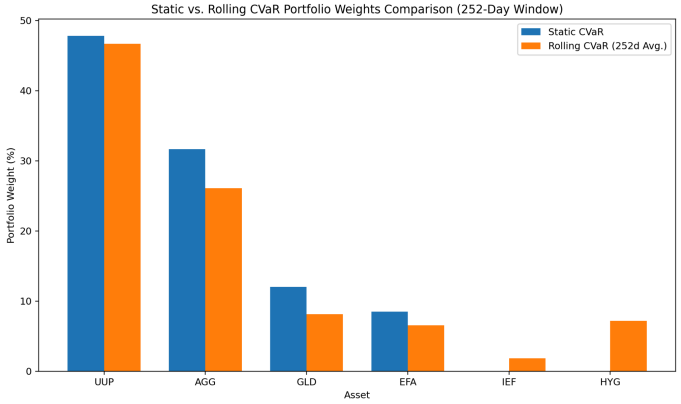
The rolling minimum-variance strategy remains highly defensive throughout the test period. On average, the portfolio is concentrated mainly in AGG (52.53%) and UUP (31.13%), with smaller allocations to HYG (9.56%), GLD (5.21%), and DBC (0.95%), while most other assets receive negligible weights. This allocation pattern is consistent with the objective of the strategy, which prioritizes variance reduction rather than return maximization.

In terms of out-of-sample performance, the rolling minimum-variance portfolio delivers a total return of 2.01%, an annualized return of 4.17%, and an annualized volatility of 2.66%. Its Sharpe ratio is 0.18, while the maximum drawdown is limited to -1.54%. The strategy also maintains relatively low tail risk, with daily VaR 95% of 0.252% and daily CVaR 95% of 0.355%. Overall, the results suggest that the rolling minimum-variance approach provides strong downside protection and stable risk control, although its return and risk-adjusted performance remain modest in this sample.

4.4 CVaR Optimization

The static CVaR portfolio produces a defensive allocation concentrated mainly in UUP, AGG, GLD, and EFA. The rolling-window CVaR strategy preserves the same defensive profile but becomes time-varying, with UUP remaining the dominant holding while secondary allocations rotate across AGG, IEF, GLD, and EFA. This pattern is consistent with the objective of the model, which prioritizes downside-risk control rather than return maximization.

In terms of out-of-sample performance, the rolling-window CVaR strategy achieves an annualized return of 7.88%, annualized volatility of 3.08%, daily VaR of 0.26%, daily CVaR of 0.43%, and a maximum drawdown of -3.67%. Mean turnover is 0.131, suggesting that the rolling-window framework improves adaptability without excessive trading activity. Overall, the results indicate that the rolling-window CVaR strategy maintains a stable defensive allocation while allowing the portfolio to respond to changing downside-risk conditions.



This figure compares the fixed-weight allocation from the static CVaR model with the average allocation under the 252-day rolling-window CVaR strategy. Both portfolios remain defensive, with UUP as the dominant holding, while the rolling-window version reduces the concentration in AGG and redistributes part of the secondary weight toward HYG and IEF.

5. Evaluation

This section compares the four strategies on a common set of out-of-sample metrics. For each strategy, we use the preferred specification reported in Section IV (static equal-weight; GARCH Dynamic MVO as the headline MVO variant; rolling-window minimum-variance; rolling-window CVaR).

Metric	Equal Weight	MVO (GARCH)*	Min Variance	CVaR
Rebalancing	None (static)	21-day rolling	21-day rolling	21-day rolling
Annualized Return	≈14.7%	7.56%	4.17%	7.88%
Annualized Volatility	≈9.93%	8.39%	2.66%	3.08%
Sharpe Ratio (annual)	≈0.66	0.442	0.18	≈1.36
Max Drawdown	-4.96%	—	-1.54%	-3.67%
Daily VaR 95%	—	—	0.252%	0.26%
Daily CVaR 95%	—	—	0.355%	0.43%
Mean Turnover	n/a	0.187	—	0.131

*Table 5.1. Out-of-sample comparison of the four strategies. *MVO column reports GARCH Dynamic MVO; Static and Rolling MVO bracket these figures (Static highest Sharpe, Rolling slightly lower).*

No single method dominates on every metric. The CVaR rolling portfolio delivers the highest annualized Sharpe (≈1.36), combining modest returns with a defensive allocation built on UUP and rotating exposures to AGG, IEF, GLD, and EFA. The equal-weight benchmark ranks second on Sharpe (≈0.66) but does so by accepting roughly three times the volatility in exchange for the highest absolute return. The minimum-variance portfolio achieves the smallest drawdown (-1.54%), and the lowest daily CVaR (0.355%), but its return (4.17%) is the lowest of the four. The mean-variance variants, despite being the most theoretically sophisticated, are dominated by simpler benchmarks on risk-adjusted return out of sample.

Implementation cost reinforces this picture: the CVaR portfolio has the lowest turnover among the dynamic methods (0.131 vs. 0.187 for GARCH MVO), so its net-of-cost ranking would only improve further.

6. Conclusion

This study compared four portfolio construction strategies on fifteen liquid ETFs over a 2021–2026 sample with an out-of-sample test period. Three findings stand out. First, no single method dominates: CVaR rolling wins on Sharpe (≈1.36), minimum-variance wins on drawdown (-1.54%) and daily CVaR (0.355%), the equal-weight benchmark delivers the highest absolute return, and the mean-variance variants are dominated by simpler benchmarks on risk-adjusted return out of sample. The appropriate choice depends on the investor's objective rather than on the sophistication of the model.

Second, parameter stability matters as much as optimization quality. Methods that depend on the expected-return vector are exposed to estimation error that overwhelms their theoretical benefits; methods that depend only on covariance (minimum-variance) or tail behavior (CVaR) are more stable out of sample. Third, implementation cost and concentration risk are first-order concerns: the lowest-turnover strategy is also the highest-Sharpe strategy in our sample, and the defensive portfolios concentrate heavily in UUP and AGG.

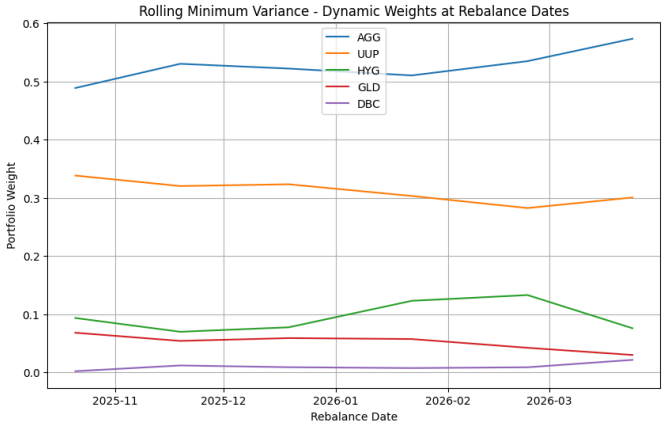
7. APPENDIX

Minimum Variance

Appendix A. Summary Performance of the Rolling Minimum-Variance Portfolio

Metric	Total Return	Annualized Return	Annualized Volatility	Sharpe Ratio	Maximum Drawdown	Daily VaR 95%	Daily CVaR 95%
Value	2.01%	4,17%	2.66%	0.18	-1.54%	0.252%	0.355%

Appendix B. Graph for rolling minimum-variance dynamic weights



Appendix C. Table for rolling minimum-variance average portfolio weights

Asset	Average Weight
AGG	0.5252686969402900
UUP	0.3113396731961950
HYG	0.0956373237367832
GLD	0.05205111375855380
DBC	0.009549418116981760
EEM	0.006153774250409740
IEF	3.60897602330059E-13
TLT	1.78755104791105E-13
IWM	1.20620092311378E-13
QQQ	9.31940373266644E-14
SPY	9.22031125341663E-14
VNQ	7.87993427970028E-14
LQD	4.03459779927675E-14
BITO	1.06393143655197E-15
EFA	1.28870096511823E-16

CVaR

Appendix A. Rolling-Window CVaR Additional Results

To supplement the main text, this appendix reports additional results for the rolling-window CVaR strategy estimated with a **252-day window** and rebalanced every 21 trading days. **Figure A1** shows the full dynamic weight path at rebalance dates. The portfolio remains anchored by UUP over most of the sample, while AGG, IEF, GLD, and EFA rotate in response to changing downside-risk conditions. **Figure A2** presents the cumulative return path of the rolling-window strategy, which shows a generally positive performance trend over the training period. **Figure A3** reports realized 95% VaR over time and shows that downside risk is time-varying rather than constant. Supporting summary statistics are reported in **Table A1**, while turnover results are reported in **Table A2**. These additional results provide supporting evidence for the main discussion in the report.

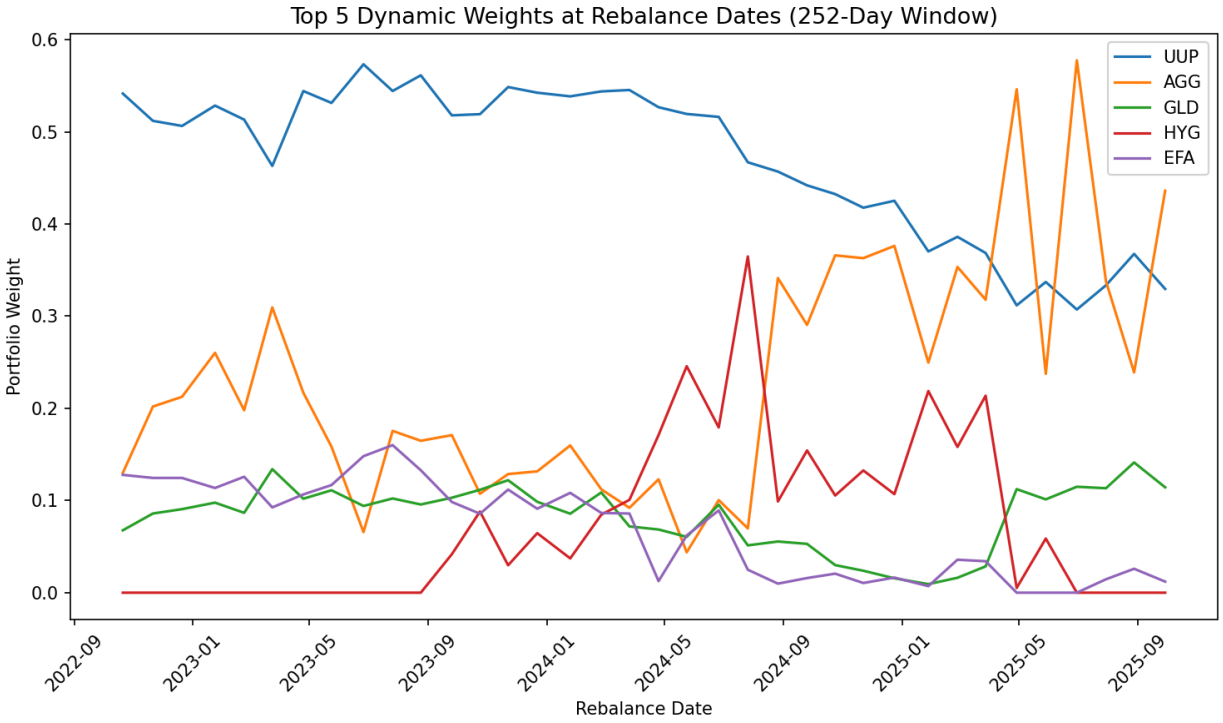


Figure A1. Rolling-Window CVaR Dynamic Weights at Rebalance Dates

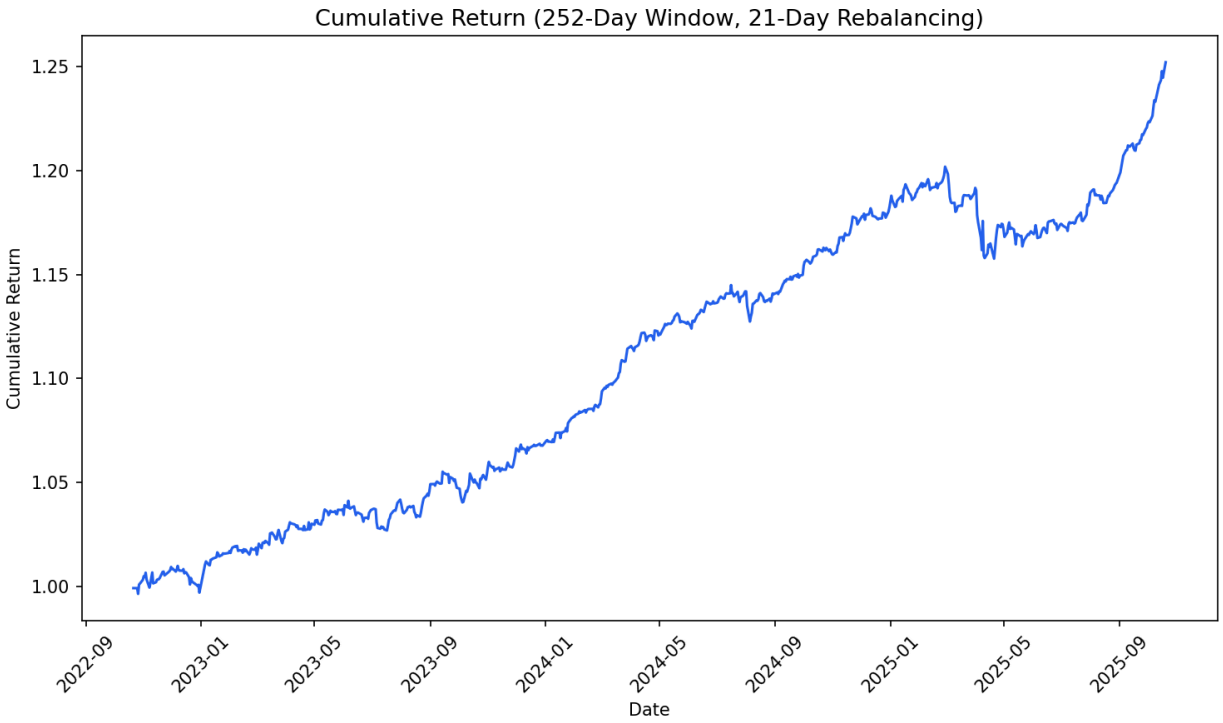


Figure A2. Rolling-Window CVaR Cumulative Return

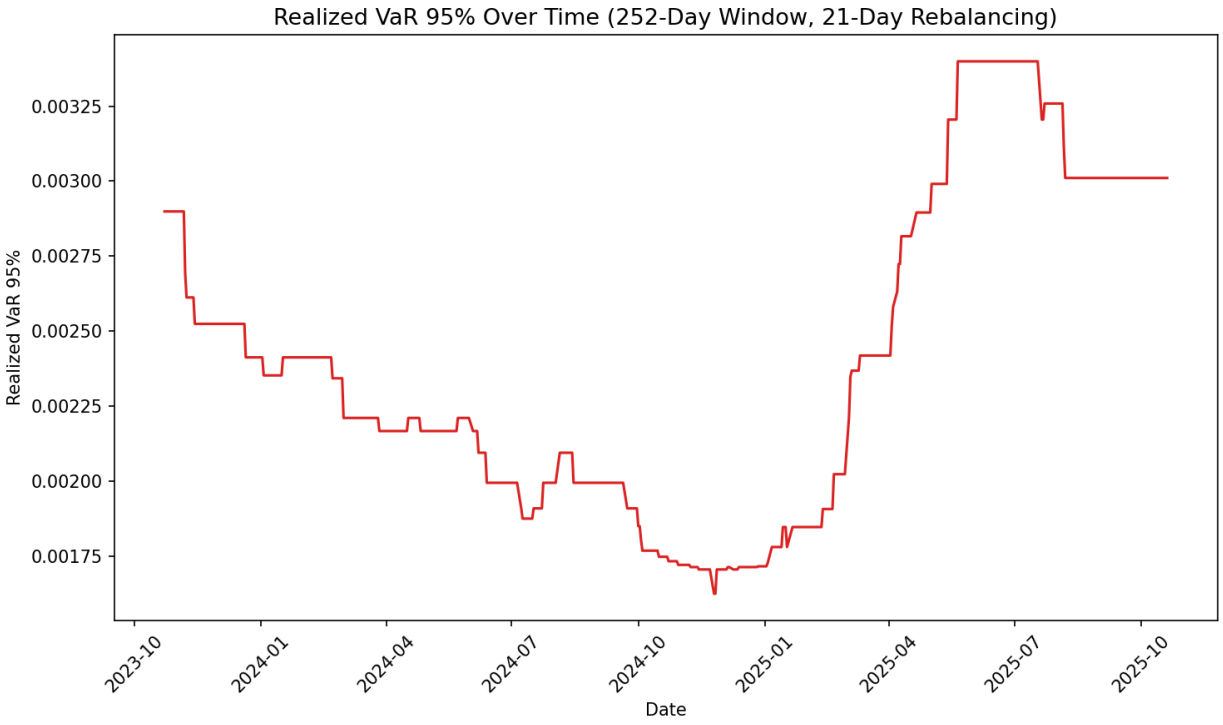


Figure A3. Rolling-Window CVaR Realized 95% VaR Over Time

rolling_cvar_summary_metrics_train_252d_r

Metric	Value
Estimation Window	252.0
Rebalance Frequency	21.0
Mean Daily Return	0.000301061676319773
Annualized Return	0.0788073488494943
Daily Volatility	0.00193961859072911
Annualized Volatility	0.0307904905763207
VaR 95% (Loss)	0.00263980141691012
CVaR 95% (Loss)	0.00432961572764840
Downside Deviation	0.00129409065428336
Semivariance	1.67467062150352E-06
Maximum Drawdown	-0.0367457249882196
Sharpe Ratio (daily)	0.0583709739978022
Sortino Ratio (daily)	0.0874880178991790

Table A1. Summary Performance and Downside-Risk Measures for the Rolling-Window CVaR Strategy

rolling_cvar_turnover_summary_train_252d

Metric	Value
Mean Turnover	0.130848741426097
Median Turnover	0.0988558583134487
Max Turnover	0.354030812746293

Table A2. Turnover Summary for the Rolling-Window CVaR Strategy

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Appendix B. Target-Return Sensitivity Analysis

This appendix also reports sensitivity results for target-return-constrained CVaR portfolios. As the required target return increases, the allocation gradually shifts away from AGG and toward UUP and GLD, while small equity exposure begins to appear at higher target levels. At the same time, daily CVaR rises, indicating that stronger return requirements come at the cost of higher downside risk. These target-return results are summarized in **Table B1**. To provide additional intuition, **Figure B1** plots the sensitivity of portfolio risk and return to changes in UUP and AGG weights. The figure suggests that UUP contributes more efficiently to downside protection, while AGG has a stronger return drag. Together, these results help explain the weight adjustments observed under target-return constraints.

Target Return	Main Weights	Daily CVaR 95%
No target	UUP 47.8%, AGG 31.7%, GLD 12.0%, EFA 8.5%	0.42%
8% target	UUP 52.0%, AGG 21.0%, GLD 17.6%, EFA 9.3%	0.44%
10% target	UUP 58.2%, GLD 24.6%, EFA 10.0%, AGG 7.2%	0.51%
12% target	UUP 57.8%, GLD 33.2%, EFA 7.0%, QQQ 1.9%	0.60%
13% target	UUP 54.8%, GLD 38.6%, SPY 6.1%	0.67%

Table B1. CVaR Portfolio Weights and Daily CVaR Under Alternative Target-Return Constraints

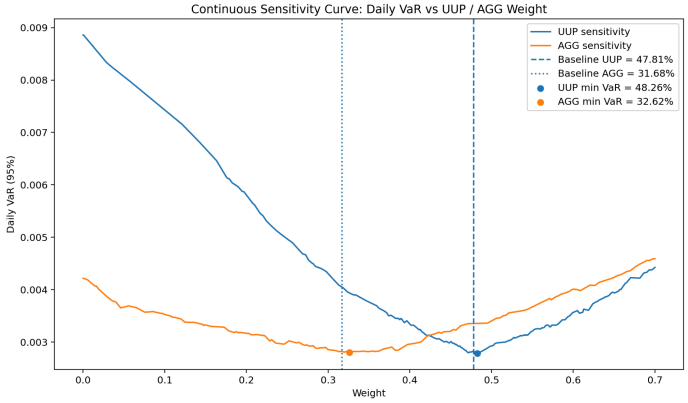


Figure B1. Sensitivity Analysis of UUP and AGG

Appendix C. Extended Evaluation Discussion

C.1 Risk-Return Trade-off (Extended)

The four strategies occupy strikingly different regions of the risk-return plane. The CVaR rolling portfolio combines a defensive allocation with consistent positive drift, producing the highest Sharpe of the four methods. The equal-weight portfolio achieves a competitive Sharpe through diversified exposure to assets that performed well during the test window, including U.S. equities and gold, but pays for this with volatility roughly three times higher than the variance-minimizing portfolios. The minimum-variance portfolio sits at the opposite extreme: the lowest volatility of any specification (2.66%) at the cost of the lowest return (4.17%). The GARCH Dynamic MVO portfolio is over-weighted toward IEF (49.3%) because GARCH improves the covariance estimate but not the mean estimate, leading the optimizer to penalize risky assets disproportionately.

C.2 Tail-Risk and Drawdown Behavior (Extended)

A notable inversion appears in the tail-risk ranking: the minimum-variance portfolio achieves a lower daily 95% CVaR (0.355%) than the CVaR-optimized portfolio itself (0.43%). This is not a contradiction. For portfolios with roughly normal daily returns, variance and CVaR are tightly correlated, so the minimum-variance portfolio inherits strong tail protection as a by-product of its variance objective. The CVaR portfolio explicitly trades a small amount of tail-risk performance for higher expected return by including UUP and rotating allocations across AGG, IEF, GLD, and EFA. The trade-off is consistent with each method's objective: investors prioritizing capital preservation should prefer minimum-variance, while investors targeting risk-adjusted return should prefer the CVaR specification.

C.3 Stability, Concentration, and Implementation Cost (Extended)

Three implementation considerations differentiate the strategies beyond raw performance.

· **Weight concentration.** The minimum-variance, CVaR, and GARCH MVO portfolios are highly concentrated in a small number of defensive assets (UUP + AGG dominating the first two; IEF dominating the third). The equal-weight portfolio is the only specification with uniform diversification across all fifteen ETFs. High concentration creates idiosyncratic exposure to a few assets and reduces the diversification benefit a multi-asset portfolio would otherwise offer.

· **Turnover.** CVaR rolling has the lowest mean turnover of the dynamic methods (0.131), versus 0.158 for Rolling MVO and 0.187 for GARCH MVO. Lower turnover translates directly into lower transaction costs in any live implementation, improving the CVaR portfolio's net-of-cost ranking further.

· **Estimation-error sensitivity.** Rolling MVO continuously re-estimates both μ and Σ, but the noise in μ dominates the benefit of adaptability. GARCH MVO improves the covariance estimate without correspondingly improving the mean estimate, so the optimizer overweights low-risk assets and sacrifices return for variance reduction. Minimum-variance and CVaR do not depend on μ at all, which is the central reason for their stronger out-of-sample stability.

Appendix D. Code for Data Collection and Processing

```python
import yfinance as yf
import pandas as pd
import numpy as np
from pandas_datareader import data as pdr
from google.colab import drive
drive.mount('/content/drive')

```

```
Mounted at /content/drive
```

```python
tickers = ["SPY","QQQ","IWM","EFA","EEM","AGG","IEF","TLT","LQD","HYG","GLD","DBC","VNQ","UUP","BITO"]

start = "2021-10-19"   # 因为包含 BITO，所以共同起点设为 BITO 上市日
end   = "2026-04-20"

# 1. 下载 Yahoo Finance 日度复权收盘价
prices = yf.download(
    tickers,
    start=start,
    end=end,
    interval="1d",
    auto_adjust=True,
    progress=False
)["Close"]

# 2. 计算资产日收益率
returns = prices.pct_change().dropna()


print(prices.head())
print(returns.head())
print(prices.shape)
```

```
Ticker            AGG       BITO        DBC        EEM        EFA         GLD  \
Date
2021-10-19  98.783623        NaN  18.794979  46.920723  69.528610  165.449997
2021-10-20  98.714417  11.674772  19.003815  46.911785  69.771606  166.899994
2021-10-21  98.506737  11.013883  18.716669  46.607922  69.441841  166.750000
2021-10-22  98.697121  10.657811  18.803682  46.509609  69.858383  167.770004
2021-10-25  98.792297  10.946447  18.899397  46.750919  69.719536  168.929993
```

```
dtb3 = pdr.DataReader("DTB3", "fred", start, end) #用dtb3来代理日度rf rate
dtb3.head()
```

DATE	DTB3
2021-10-19	0.05
2021-10-20	0.05
2021-10-21	0.06
2021-10-22	0.06
2021-10-25	0.06

```
dtb3 = dtb3.rename(columns={"DTB3": "dtb3_annual_pct"})
dtb3["rf_annual"] = dtb3["dtb3_annual_pct"] / 100
dtb3_aligned = dtb3.reindex(returns.index)
dtb3_aligned["rf_annual"] = dtb3_aligned["rf_annual"].ffill()
dtb3_aligned["rf_daily"] = dtb3_aligned["rf_annual"] / 252
```

```
data = returns.join(dtb3_aligned[["rf_daily"]], how="left")
cols = ["rf_daily"] + [c for c in data.columns if c != "rf_daily"]
data = data[cols]
data=data.rename(columns={"rf_daily": "rf_rate_daily"})
```

```
train = data.loc["2021-10-21":"2025-10-20"].copy()
test  = data.loc["2025-10-21":"2026-04-20"].copy()

print("train shape:", train.shape)
print("test shape:", test.shape)
print(train.head())
print(test.head())
```

```
train shape: (1003, 16)
test shape: (123, 16)
            rf_rate_daily       AGG      BITO       DBC       EEM       EFA  \
Date
2021-10-21       0.000002 -0.002104 -0.056608 -0.015110 -0.006477 -0.004726
2021-10-22       0.000002  0.001933 -0.032329  0.004649 -0.002109  0.005998
2021-10-25       0.000002  0.000964  0.027082  0.005090  0.005188 -0.001988
2021-10-26       0.000002  0.001839 -0.012814  0.001381 -0.005162  0.001992
2021-10-27       0.000002  0.003322 -0.049925 -0.018391 -0.008071 -0.003975

                 GLD       HYG       IEF       IWM       LQD       QQQ  \
Date
2021-10-21 -0.000899 -0.002521 -0.002632  0.002551 -0.002120  0.006107
2021-10-22  0.006117 -0.002412  0.002551 -0.002369  0.004022 -0.008402
2021-10-25  0.006914  0.001728  0.001228  0.009498  0.000453  0.010238
2021-10-26 -0.007400  0.000229  0.001227 -0.007536  0.004306  0.003149
2021-10-27  0.002624 -0.000805  0.005602 -0.018390  0.006093  0.002321

                 SPY       TLT       UUP       VNQ
Date
2021-10-21  0.002608 -0.001191  0.002390  0.001384
2021-10-22 -0.001036  0.011013 -0.001987  0.004699
2021-10-25  0.005363 -0.001526  0.001592  0.003210
2021-10-26  0.000900  0.008269  0.001987  0.002742
2021-10-27 -0.004430  0.018194 -0.000397 -0.008479
            rf_rate_daily       AGG      BITO       DBC       EEM       EFA  \
Date
2025-10-21       0.000151  0.000989  0.008758 -0.010295 -0.011107 -0.007346
2025-10-22       0.000151 -0.000099 -0.036354  0.015830 -0.000921 -0.001480
2025-10-23       0.000151 -0.001778  0.022523  0.016919  0.008846  0.004659
2025-10-24       0.000150  0.000990  0.002753  0.000438  0.006577  0.001581
2025-10-27       0.000148  0.000692  0.039539 -0.002188  0.011071  0.006944

                 GLD       HYG       IEF       IWM       LQD       QQQ  \
Date
2025-10-21 -0.064269 -0.001484  0.001846 -0.004715  0.000178 -0.000261
2025-10-22  0.000106 -0.000991  0.000205 -0.014778  0.000977 -0.009634
2025-10-23  0.004002  0.002231 -0.003275  0.012657 -0.001243  0.008406
2025-10-24 -0.003353  0.002721  0.000924  0.012215  0.001688  0.010678
2025-10-27 -0.027840  0.002467  0.000000  0.003488  0.001686  0.017809
```